

Diverging #1

Goal: Think of as many concepts as possible without constraints (any wild, unrealistic, futuristic concepts were accepted and encouraged)

Diverging Tools Used:

1. Reverse Brainstorming

Pros:

- More creativity
- Helped us better recognize and define the basic requirements of the vending machine, that we would have looked past before
- I believe this is the diverging technique that left us with the most useable concepts in the later diverging stages (as shown in diverging #3)

Cons:

- Some ideas were out there (so a bit distracted, but mostly fine)

2. Lotus Blossom Technique

Pros:

- identify the many variables affecting accessibility of the vending machine
- Helped us branch out

Cons:

- We are still mostly focused on our constraints from our design brief (not much creativity)

3. Wishing

Pros:

- More creativity: helped us to think beyond what was possible
- Expanded our mindset → view things from a different perspective

Cons:

- Led to too many unrealistic concepts (although it was fun)

4. Challenging Assumptions

Pros:

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Cons:

- Got too off track at this point (our group has a tendency to think creatively or crazily, especially after the first three diverging techniques, so this got us too off track)

5. Random Input

Pros:

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Cons:

- The words that were generated gave no relation to the vending machine (such as "marriage")

Concepts: [attach pictures]

Unique solutions from this process:

- Teleporting snacks
- Drones to deliver snacks
- Igloo to place snacks
- meatballs?

Potential questions to answer for this stage of diverging/converging:

- Why did we choose these diverging tools?
- Which diverging tools was the most useful for this first diverging?
- What was the key advantage this diverging tool provided over the others?

Converging #1

Goal: Remove unrealistic concepts (reality filter)

Reason for removal of concepts:

- They were unrealistic or infeasible

Removed Concepts:

- Teleporting snacks
- Drones to deliver snacks
- Igloo to place snacks
- Meatballs?
- Any requirements on the user (such as: change of mindset, practice self-control, or grow taller when snack cannot be retrieved)

Potential questions to answer for this stage of diverging/converging:

Diverging #2

Goal: Build on some of the remaining realistic ideas

Diverging tools used:

- Morph chart
Purpose: functional breakdown of the problem into sub-functions
[insert picture]
- Diverging tools from diverging #1 (as some ideas were remained)

Concepts:

- Any combination of ideas in the picture
- Realistic concepts from diverging #1

Potential questions to answer for this stage of diverging/converging:

Converging #2

Goal: cut down to relating to the scope

Reason for removal of concepts:

- They altered too much of the machine itself
 - Define too much alteration: concepts that alters the anywhere above the snack placement section. This includes concepts relating to:
 - no drop of the snack to reach the bottom of opening
 - Different payment options
 - Ways to selecting the food product

Concepts remaining:

- Drawer (challenging assumption / morph chart)
- No flap, only opening (reverse brainstorming / morph chart)
- Don't have to put hand in (reverse brainstorming)
- Auto detector (reverse brainstorming / morph chart)
- Drop snack out of the machine (reverse brainstorming)
- Have robot tray to come out and deliver snacks (reverse brainstorming)
- Deliver the snack to you (reverse brainstorming)
- Sliding door (wishing)
- Doesn't have to touch anything other than snack (wishing)
- Portable device to get snacks (wishing)
- Claw (morph chart)

Potential questions to answer for this stage of diverging/converging:

Diverging #3

Goal: Build on the remaining feasible ideas with a focus on low-modification or additive assistive design concepts

Supported by information from lecture

"Creativity loves constraints"

Perceived Potential for creativity model: Bell curve: there is a "the sweet spot" between "underconstraint" (where there is a paradox of choice) and "over constraint" that are best for creativity

Diverging tools used:

- Morph chart

Purpose: functional breakdown of the problem into sub-functions such as:

- Access opening mechanism
- Snack elevation/delivery (user bending+ lifting,)
- Force reduction method
- User movement required (push or pull, wave (or just have a hand there), pinch an object, foot pedal)

10 Concepts:

- 1.

Potential questions to answer for this stage of diverging/converging:

$n + 1$ Concepts

Goal: remove too complex or high-modification required concepts

Reason/process for removal of concepts:

- Removal through scoring method (based on requirements and ECs)
 - No alteration to original machine = 1 point
 - Alteration to original machine = 0 points
 - No bending = 2 points
 - Some bending = 1 point
 - Require the same amount of bending as original or has minimal improvement = 0 points
 - ...
- Higher scores means minimal alterations to the original machine + improves user accessibility by reducing + ...

6 final concepts:

1. Having a flap inside the vending machine that allows the snacks to go out, but not allow hands to go in (Alina can attach her drawing here)

Potential questions to answer for this stage of diverging/converging: